Session 11: Sequences, Language Models and SignalP

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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A gradient is multiplied by 0.8 at each of 60 steps back along a sequence. What happens to it?

A It stays about the same

B It grows very large

C It all but vanishes: about one millionth is left

D It becomes exactly 0.8

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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In self-attention, one position gives a weight to every position. What is true of these weights?

A They are all equal

B They are 0 or 1

C They add up to 1

D They depend only on the distance

Answer now
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quiz.bioinfo.se/live

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What does a signal peptide at the N-terminus of a protein do?

A It starts translation

B It sends the protein into the secretory pathway

C It marks the protein for degradation

D It binds the protein to DNA

Answer now
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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Remembering context

RNN, LSTM, bidirectional

Learning without labels

language models, attention, Transformer

Pretraining pays off

SignalP, ESM-2 embeddings

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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What you want to learn today

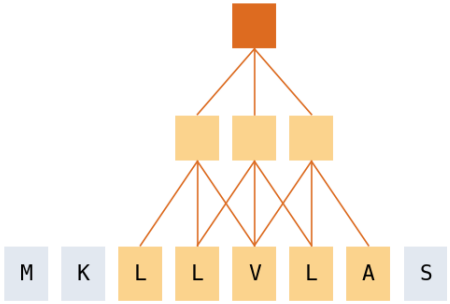
What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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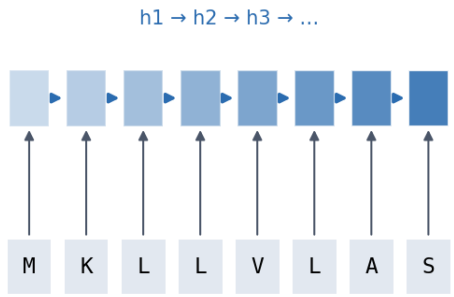
How can a model use context from the rest of a sequence?

CNN: local context grows layer by layer



a unit sees only its receptive field

RNN: a hidden state carries context



context must pass through every step in between

Attention: every position consults every other



one step, any distance; weights say how much each matters

CNN: local context grows layer by layer (Session 10). RNN: a hidden state carries context step by step. Attention: every position consults every other directly. And: how do we learn representations when labels are scarce but sequences abundant?

See course.bioinfo.se/session11.html, 'Learning goals'.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: recurrent networks and residue labelling. Block 2: language models, attention and the Transformer. Block 3: SignalP, and why pretraining helps most when labels are scarce.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

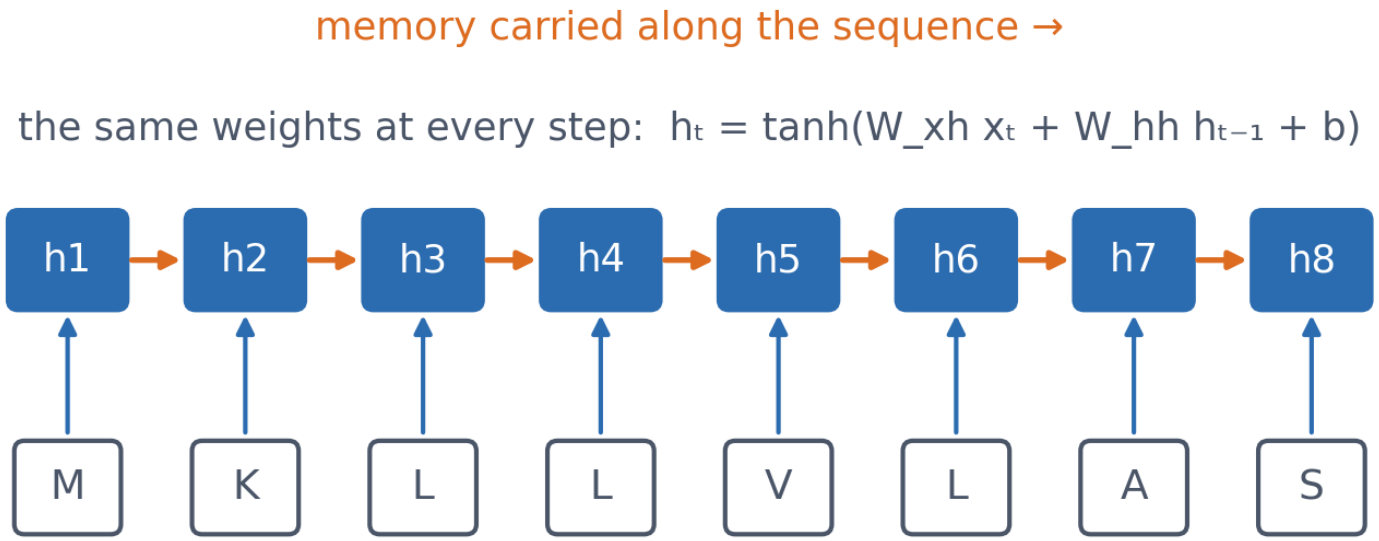
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BLOCK 1 · ABOUT 30 MIN

How can a model remember sequence context?

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A hidden state carried along the sequence

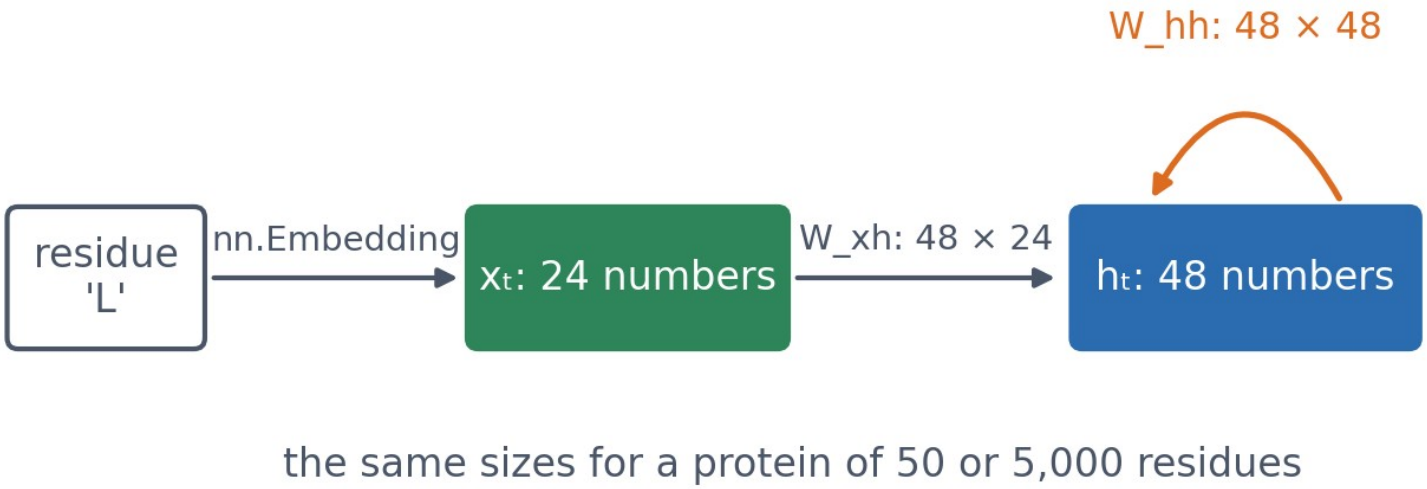


New memory = f(current residue, old memory), with the same weights at every position. One network handles proteins of any length.

See course.bioinfo.se/session11.html, 'A hidden state carried along the sequence'.

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The notebook's RNN in numbers

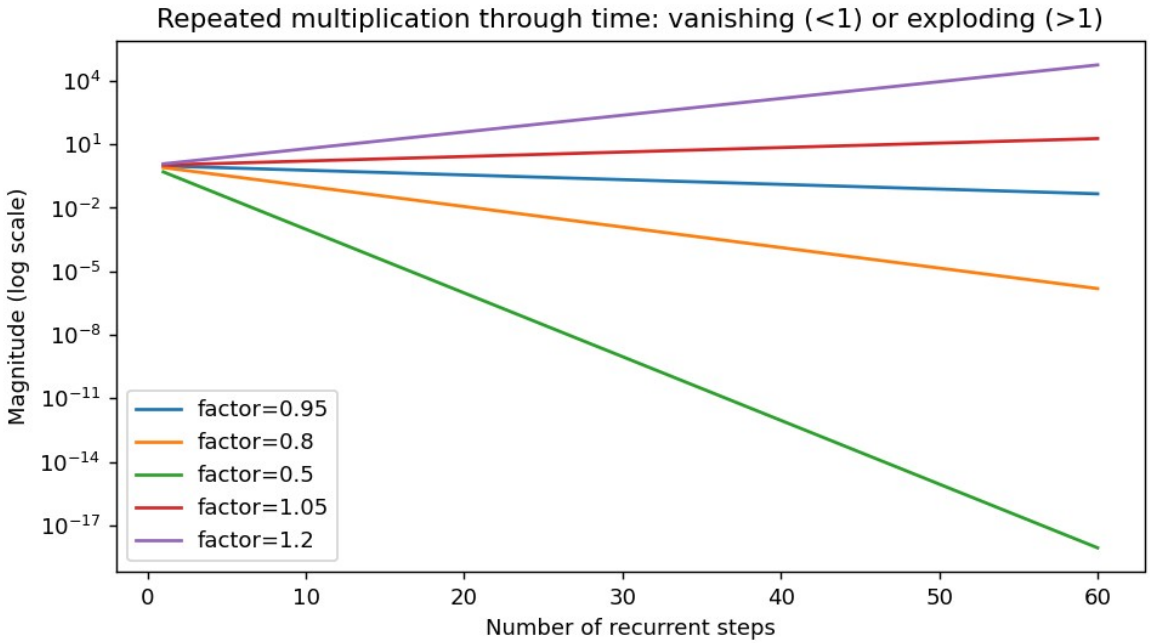


Each residue enters as a learned embedding of 24 numbers (a trainable vector per amino acid instead of a one-hot vector); the memory has 48 numbers.

Notebook: session11-rnns-lstm-gru-in-code.ipynb, Part 1.

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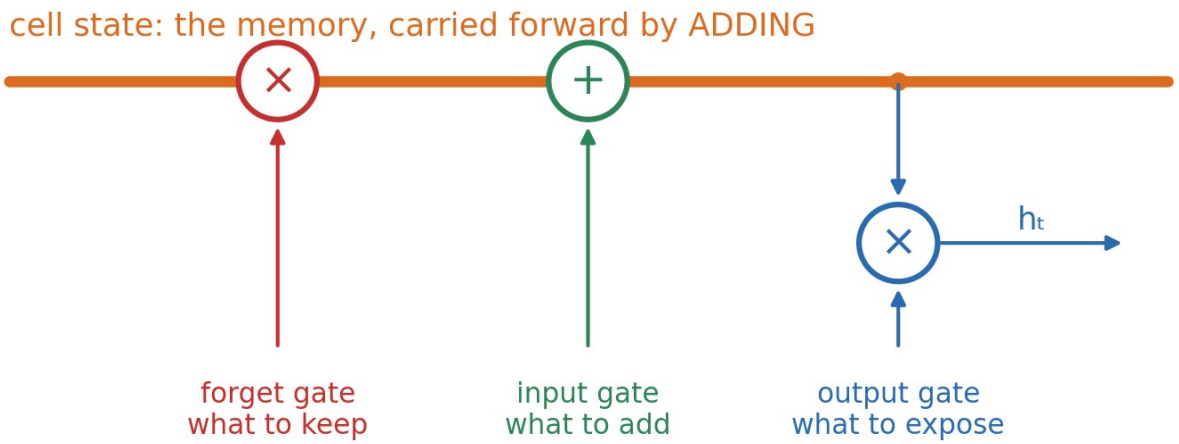
Backpropagation through time: vanish or explode



- **One factor per step**
- ×0.8 for 60 steps: 1.5×10^{-6}
- ×1.2 for 60 steps: 5.6×10^{4}
- Session 9, along the sequence

Notebook, Part 1.

LSTM: an additive memory path



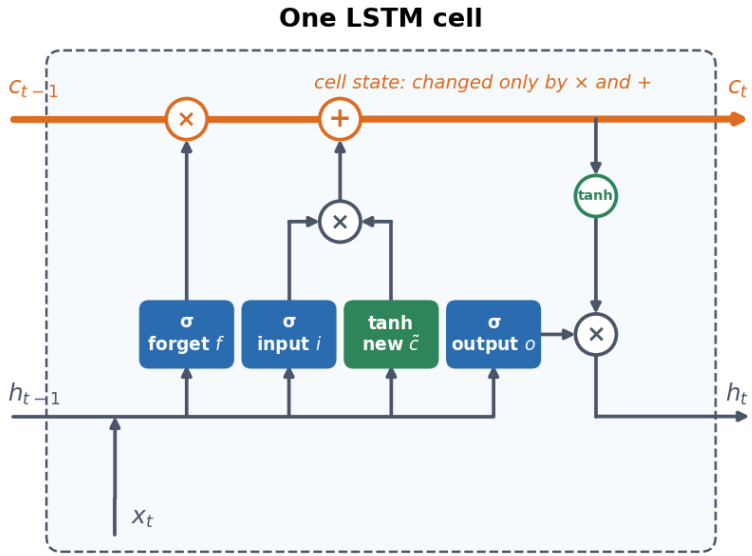
each gate: a small sigmoid network of x_t and h_{t-1}

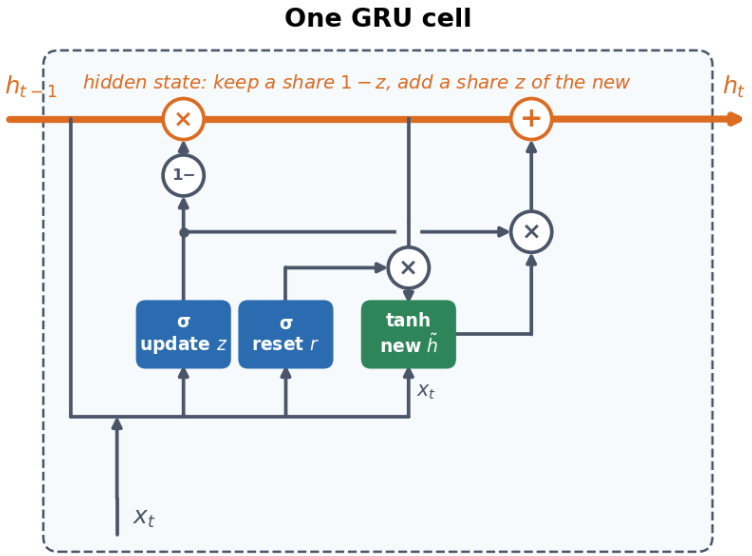
- **Memory updated by adding**
- Gates: keep, add, expose
- Error flows back over many steps
- GRU: fewer gates, no cell state

Hochreiter & Schmidhuber (1997), Neural Computation 9:1735.

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Inside an LSTM cell and a GRU cell





$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \qquad h_t = o_t \odot \tanh(c_t)$$

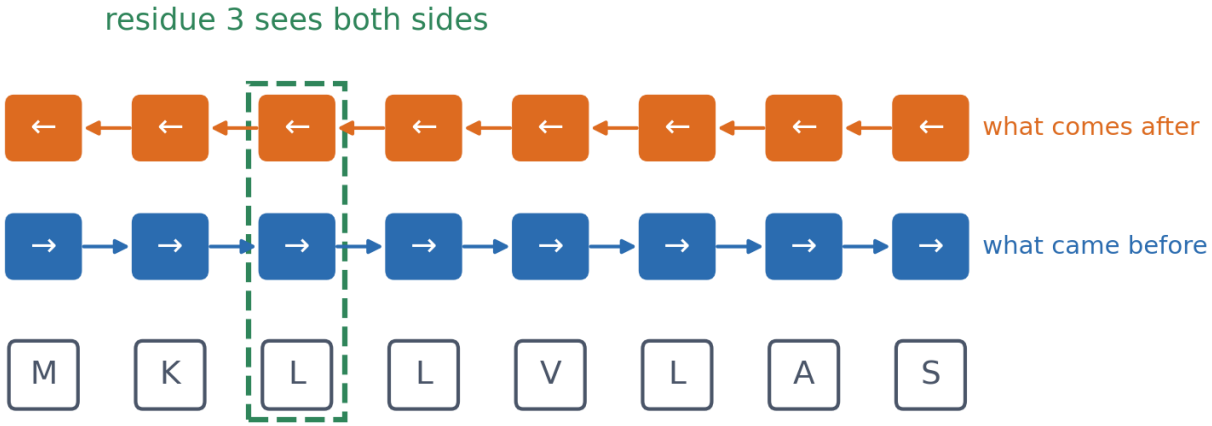
$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

Blue: sigmoid gates (0 to 1). Green: a new candidate value. LSTM: the cell state (orange) is changed only by × and +, the additive memory path. GRU: no cell state; the update gate z mixes old and new, the reset gate r decides how much of the old state the candidate sees.

Hochreiter & Schmidhuber (1997); Cho et al. (2014).

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Bidirectional: read the sequence both ways



- **One LSTM left to right**
- **One LSTM right to left**
- Combined at every position

Schuster & Paliwal (1997).

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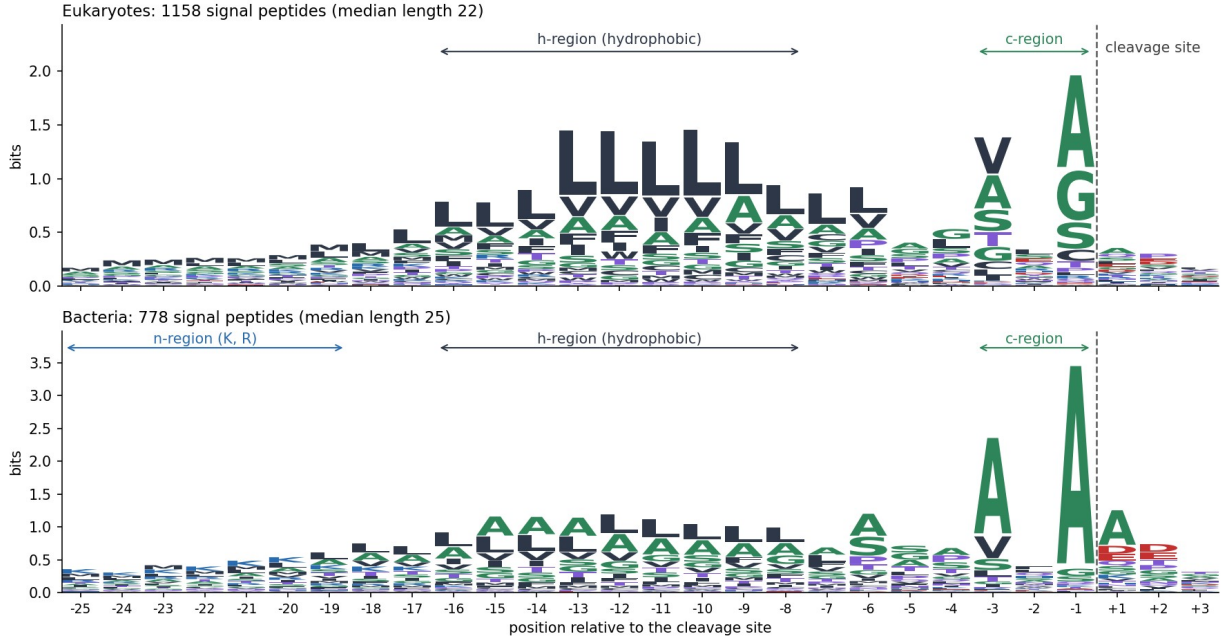
Which residues belong to a signal peptide?



- **Sends a protein to the secretory pathway**
- Three regions, then a cleavage site
- Blobel: Nobel Prize 1999

See course.bioinfo.se/session11.html, 'Case study'.

The regions are visible in the data

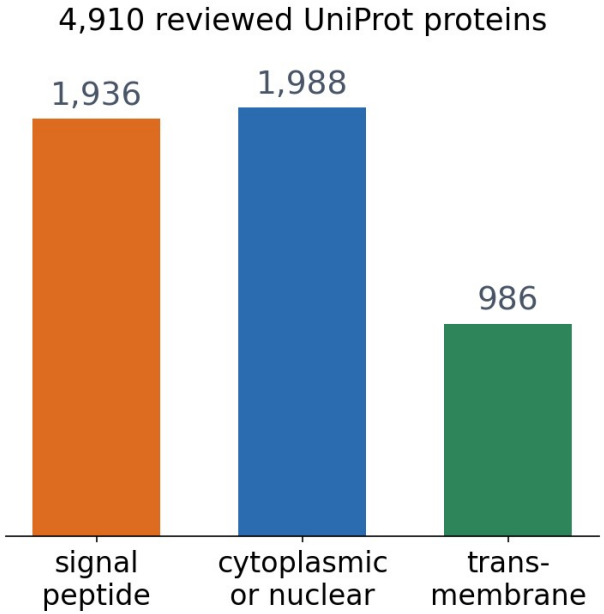


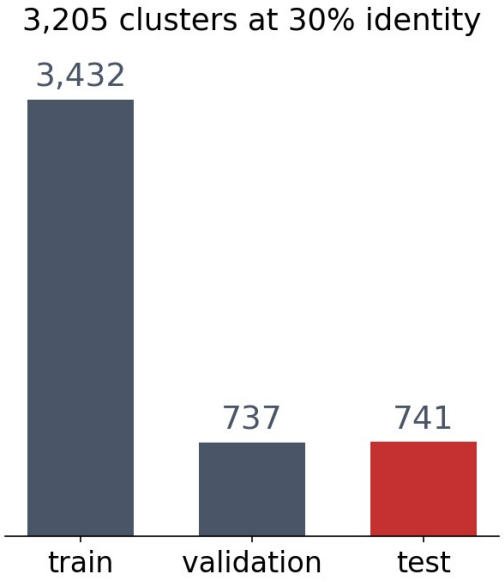
Experimentally supported signal peptides aligned on the cleavage site (dashed). Leucines fill the h-region; small residues at −3 and −1 mark the cut (the −3, −1 rule); in bacteria the n-region is rich in K and R. The two logos have different y-scales.

Our dataset: 1,158 eukaryotic and 778 bacterial signal peptides (ECO:0000269).

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The data: 4,910 proteins, homology-aware split



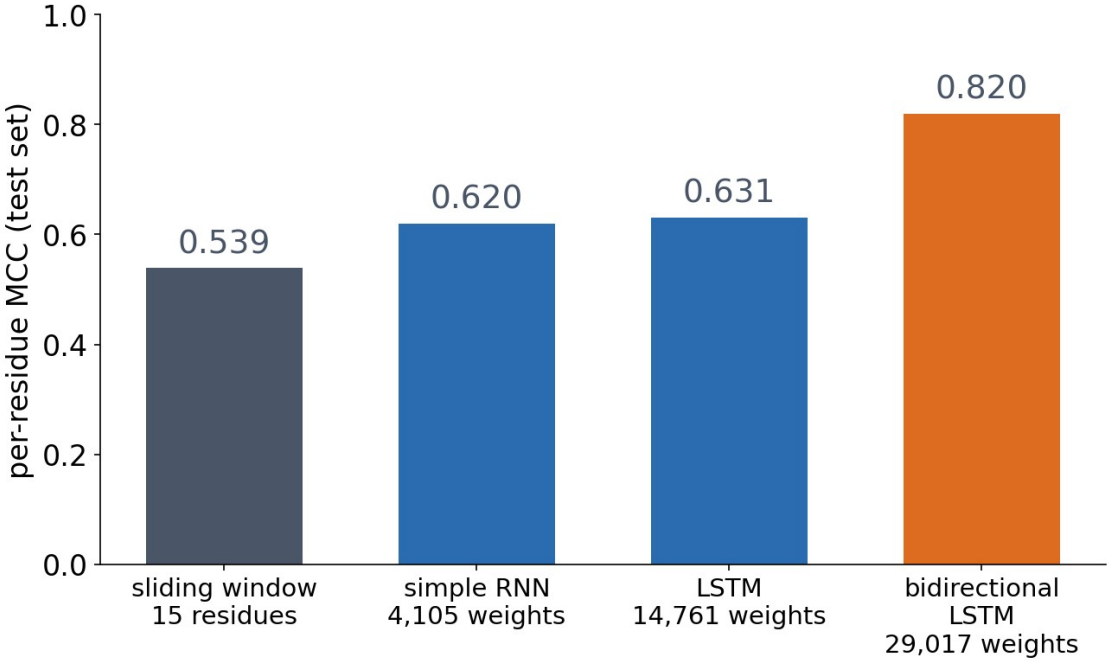


- **Experimentally supported signal peptides only**
- Transmembrane: hard negatives
- Split by MMseqs2 clusters (Session 8)

Notebook, Part 1.

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Bidirectional wins clearly

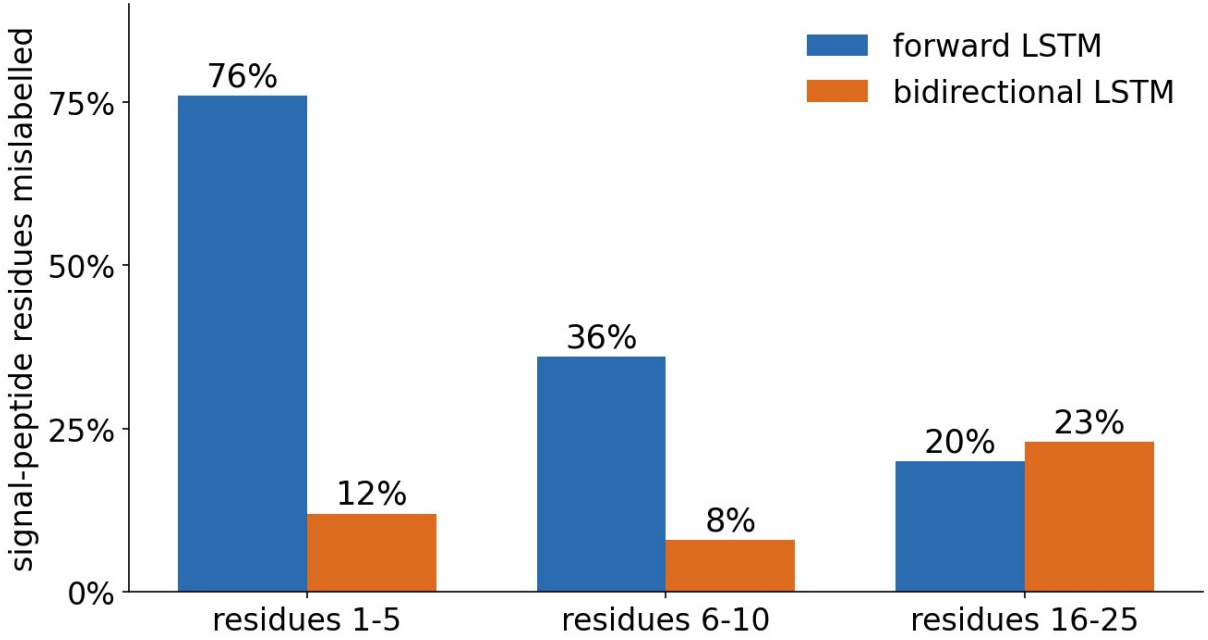


- **BiLSTM: MCC 0.820**
- Forward LSTM: 0.631
- More memory alone helps little

Notebook, Part 1; homology-aware test set.

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The evidence comes after the residue



- **Residue 3 has seen only M-x-x**
- The h-region comes later
- Near the cleavage site: a tie

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SignalP 5.0: BiLSTM + CRF



✓ starts at residue 1, one piece



✗ starts late, stops and restarts

S = signal peptide, O = other

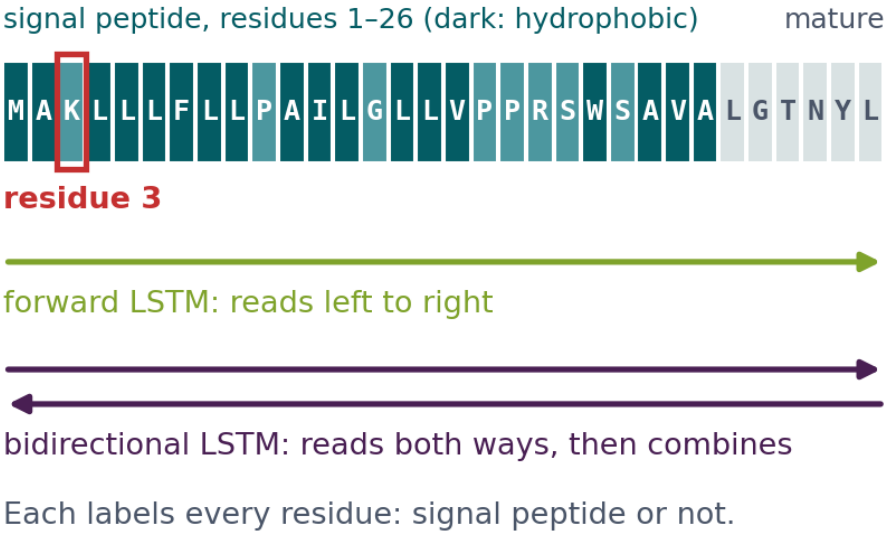
- **CRF: a structured output layer**
- Favours plausible label sequences

Almagro Armenteros et al. (2019), Nat Biotechnol 37:420.

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Where is the gap between the two LSTMs largest?

Forward and bidirectional LSTMs label each residue: signal peptide or not.



A At residues 1–5: the h-region is not yet seen

B Near the cleavage site: the hardest residues

C Evenly: bidirectional is better everywhere

D Nowhere: memory makes direction irrelevant

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Where is the gap between the two LSTMs largest?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Where is the gap between the two LSTMs largest?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session11-discussion-1.ipynb

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Where is the gap between the two LSTMs largest?

Answer again

Same question, after the discussion.



quiz.bioinfo.se/live

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Where is the gap between the two LSTMs largest?

Results, and who explains

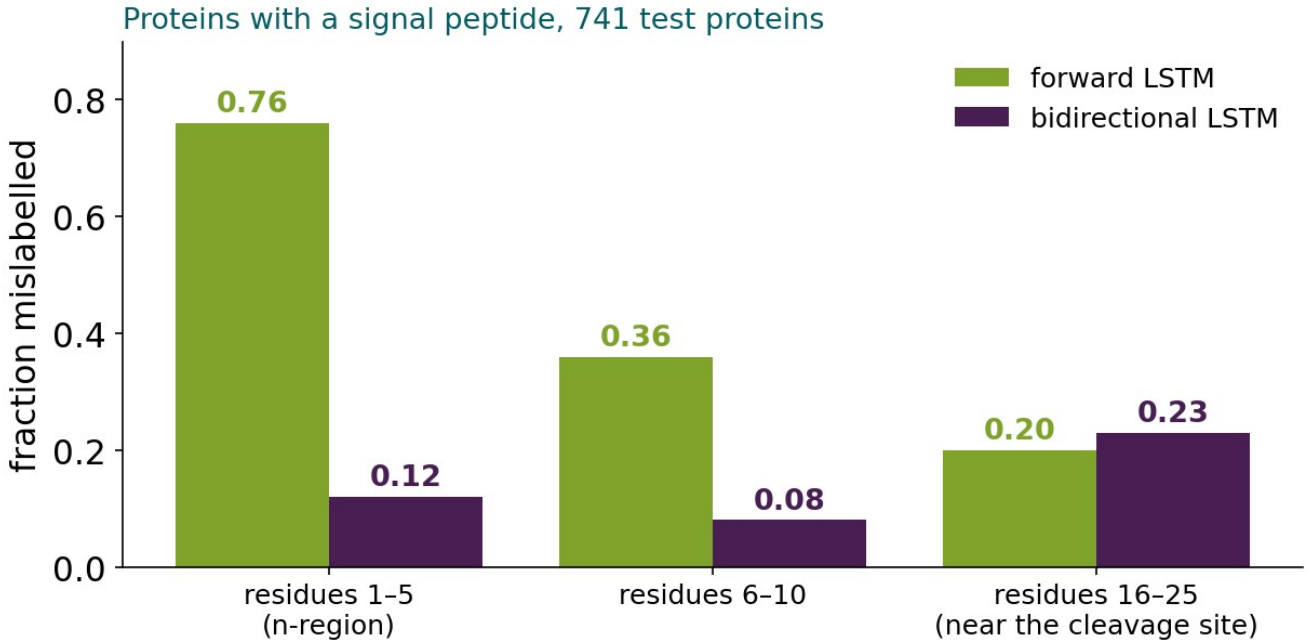
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: at the first residues, before the h-region



Whether residue 3 belongs to a signal peptide is decided by the hydrophobic h-region that comes after it. The forward LSTM has not seen it yet and mislabels three quarters of residues 1–5; the bidirectional one has. Near the cleavage site both have seen it, and both are about equally good.

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BLOCK 1 DONE

Break

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BLOCK 2 • ABOUT 30 MIN

Can we learn from sequences without labels?

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Labels are scarce; sequences are abundant

Labels

expensive

- Need experiments
- Our dataset: 1,936 experimentally supported signal peptides

Sequences

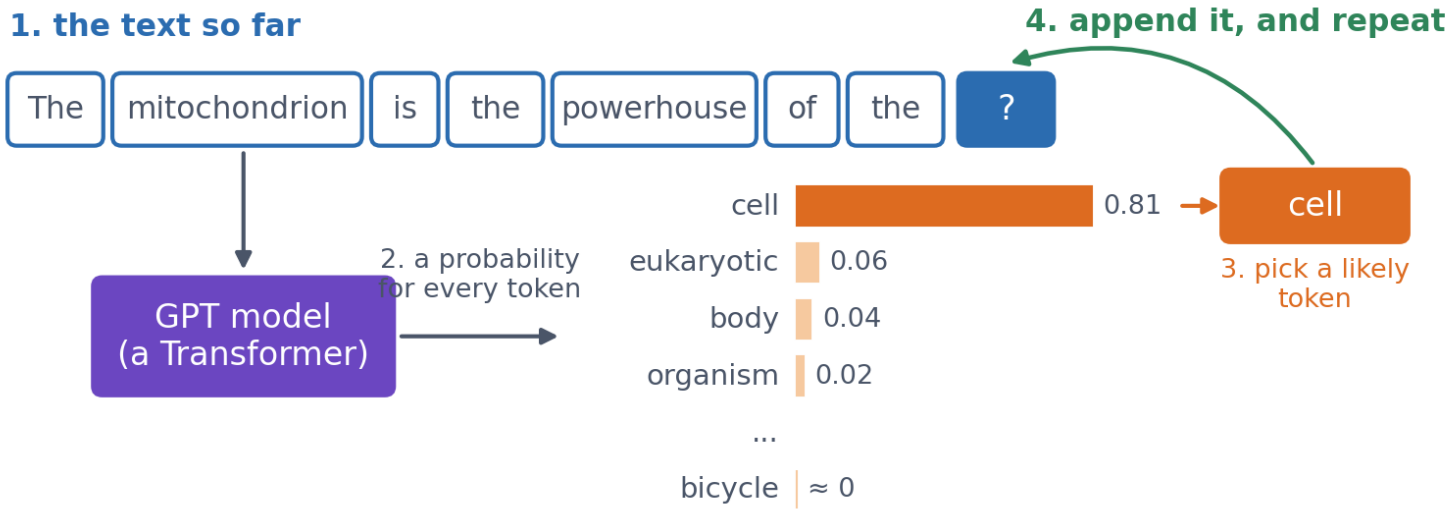
hundreds of millions

- In UniProt (Session 3)
- Free, but unlabelled

See course.bioinfo.se/session11.html, 'Can we learn from sequences without labels?'

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You have used one: how ChatGPT writes



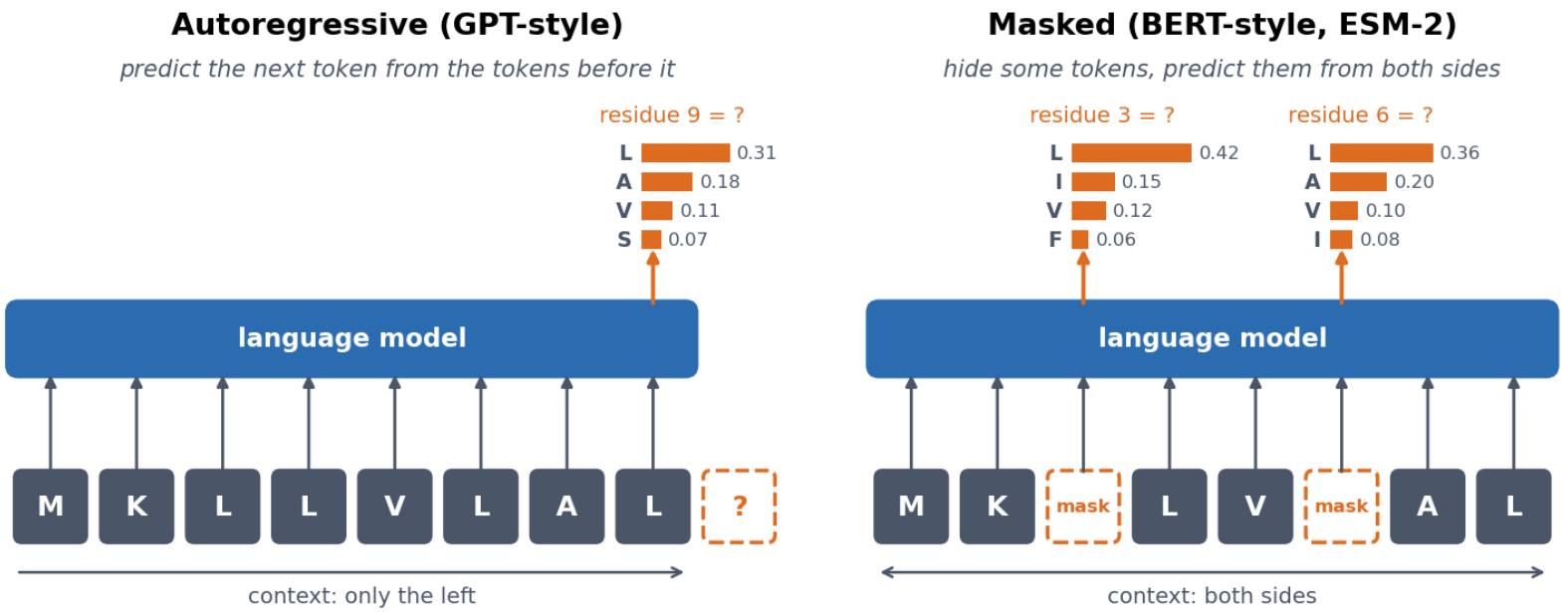
Every answer ChatGPT gives is written this way, one token at a time.

A language model gives a probability for every possible next token. GPT = Generative Pre-trained Transformer: to write, it predicts, picks a likely token, appends it and repeats. To predict well it must learn grammar, facts and arguments; for proteins, biochemistry.

Schematic; the probabilities are illustrative. See course.bioinfo.se/session11.html, 'What a language model is'.

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Two ways to learn without labels

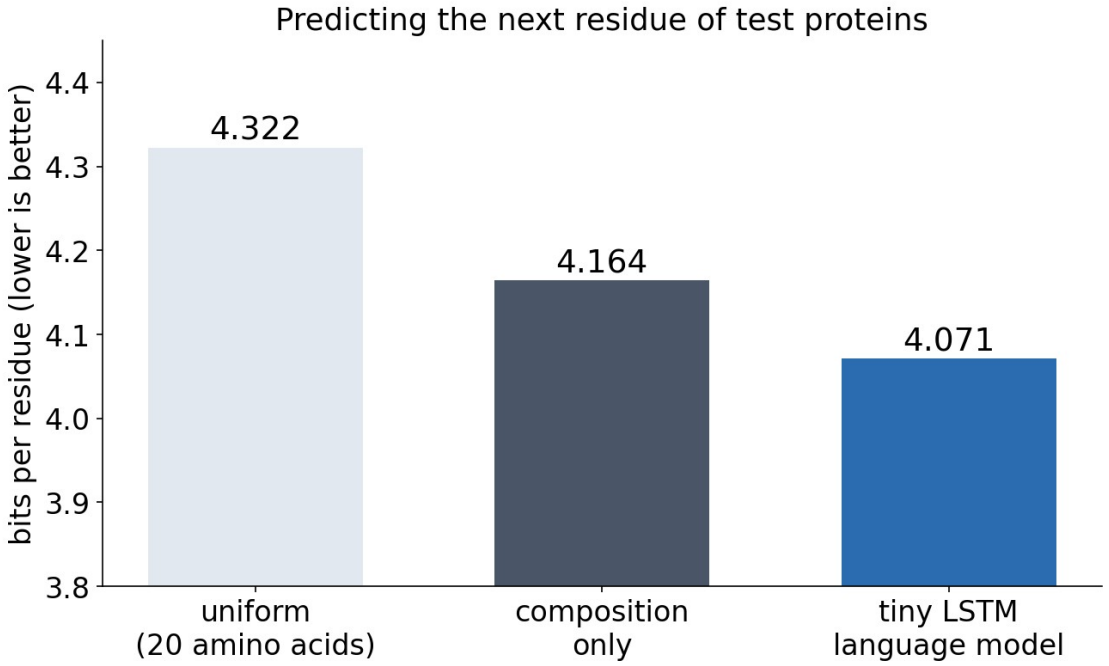


Autoregressive (GPT-style): predict the next token from the previous ones. Masked (BERT-style, ESM-2): hide some tokens and predict them from both sides.

See course.bioinfo.se/session11.html, 'What a language model is'.

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A tiny language model in the notebook

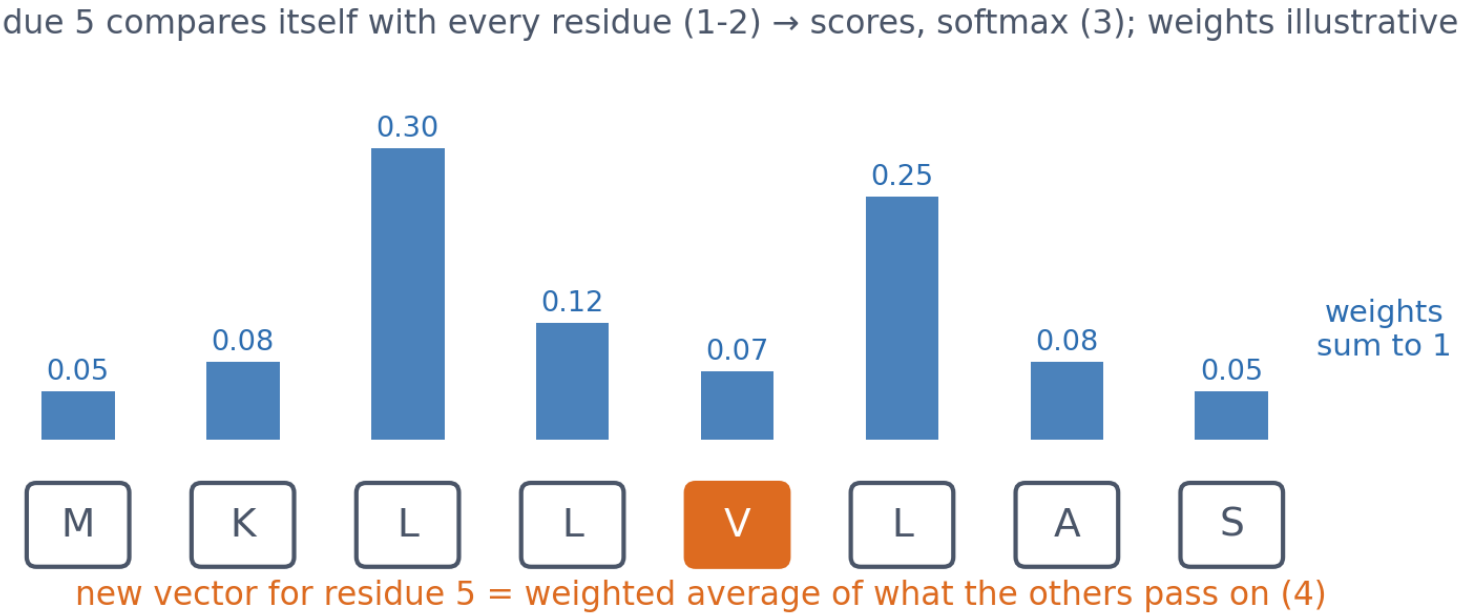


- **Next-residue LSTM: 4.071 bits**
- Composition: 4.164 bits
- Only a little better
- Large models: far more data, better architecture

Notebook, Part 2.

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Self-attention: which positions should I listen to?

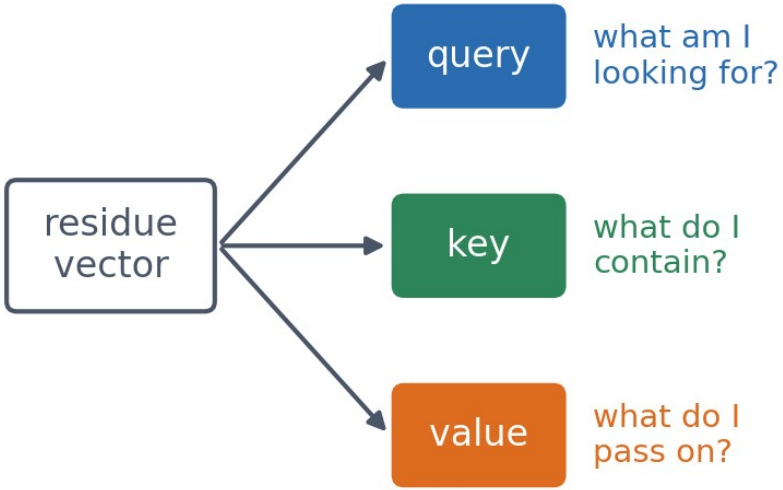


For residue i: (1) compare with every residue j, (2) give each a relevance score, (3) normalise the scores with softmax so they sum to 1, (4) update i with a weighted average.

See course.bioinfo.se/session11.html, 'Attention, from scratch'.

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Query, key and value



$$\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

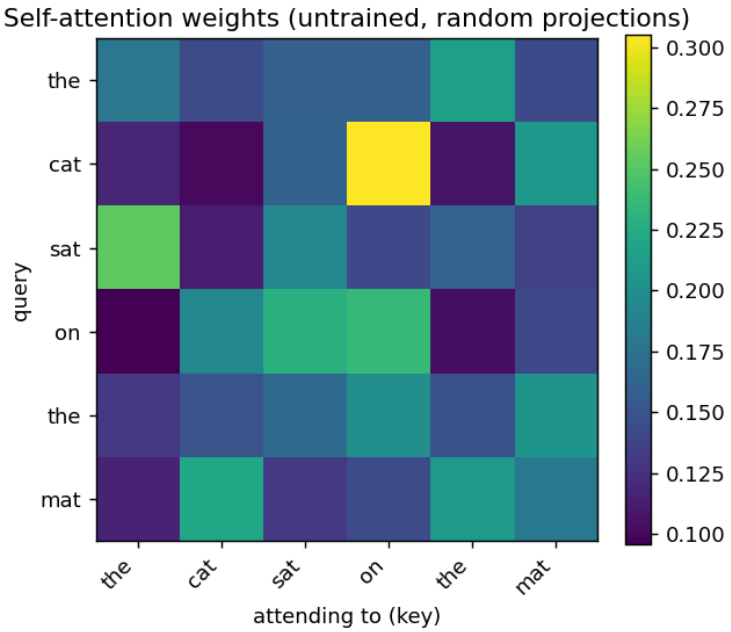
query of i · key of j = score

Three learned linear maps turn each residue's vector into a query, a key and a value. The score of j for i is the match between the query of i and the key of j.

Vaswani et al. (2017); 3Blue1Brown ch. 6 is embedded on the page.

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Any two positions, in one step



- **Each row sums to 1**
- Any distance, one step
- All positions in parallel
- **High weight ≠ contact, ≠ explanation**

Notebook, Part 2 (random, untrained projections).

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Attention alone has no sense of order

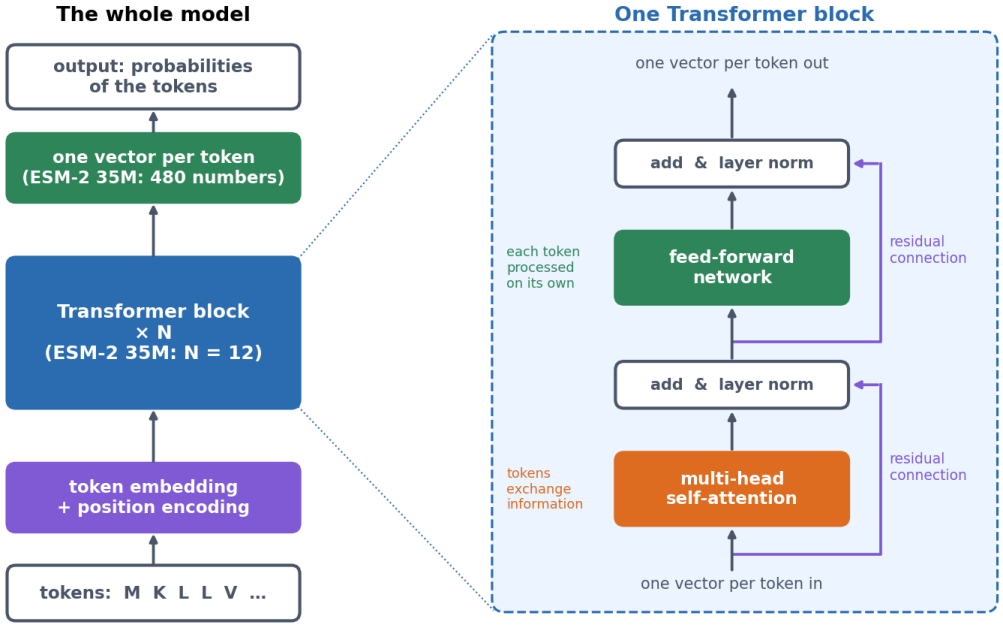


Shuffle the input and the outputs are only shuffled: 'ACDE' and 'EDCA' give the same set of vectors. The Transformer adds position encoding.

See course.bioinfo.se/session11.html, 'Attention, from scratch'.

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The Transformer: attention plus four tricks



Several heads · a feed-forward network per token (Session 7's MLP) · residual connections and layer normalisation (Session 10's ResNet) · position encoding.

Vaswani et al. (2017); 3Blue1Brown ch. 5 is embedded on the page.

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ESM-2 in the notebook

- **A stack of identical blocks**
- One vector per residue in and out
- Largest ESM-2: 48 blocks, 15 billion weights

12
blocks

20
attention heads

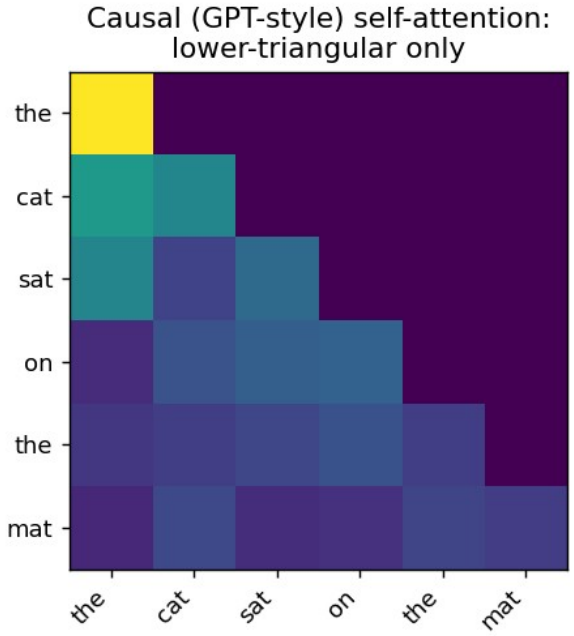
480
numbers per residue

33.5M
weights

See course.bioinfo.se/session11.html, 'The Transformer: attention plus four tricks'.

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BERT vs GPT is one mask

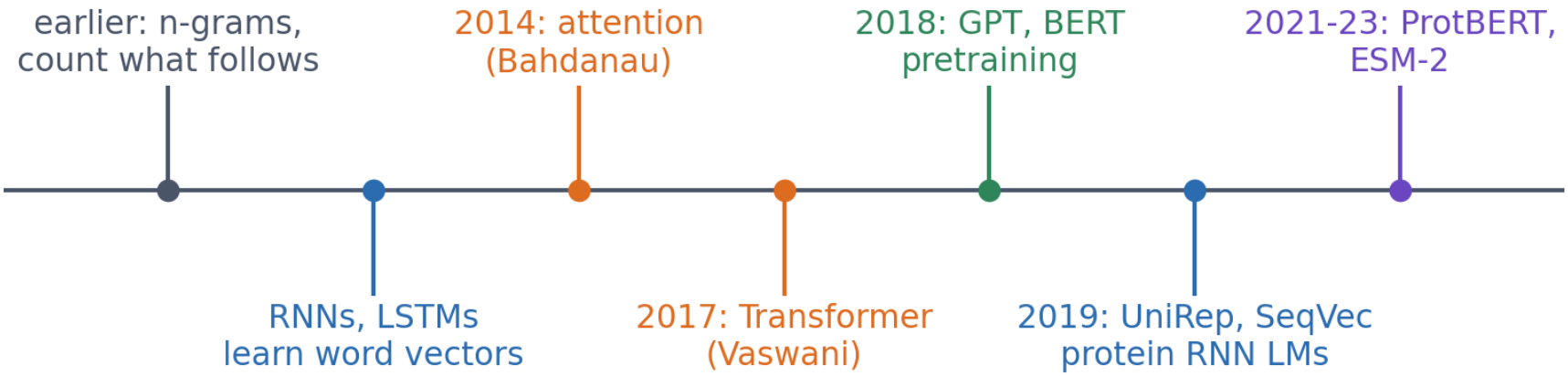


- **GPT: mask the future**
- Can generate sequences
- **BERT, ESM-2: see both sides**
- Fine for annotation

Notebook, Part 2: the weight on any future position is exactly 0.0.

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From counting words to protein language models

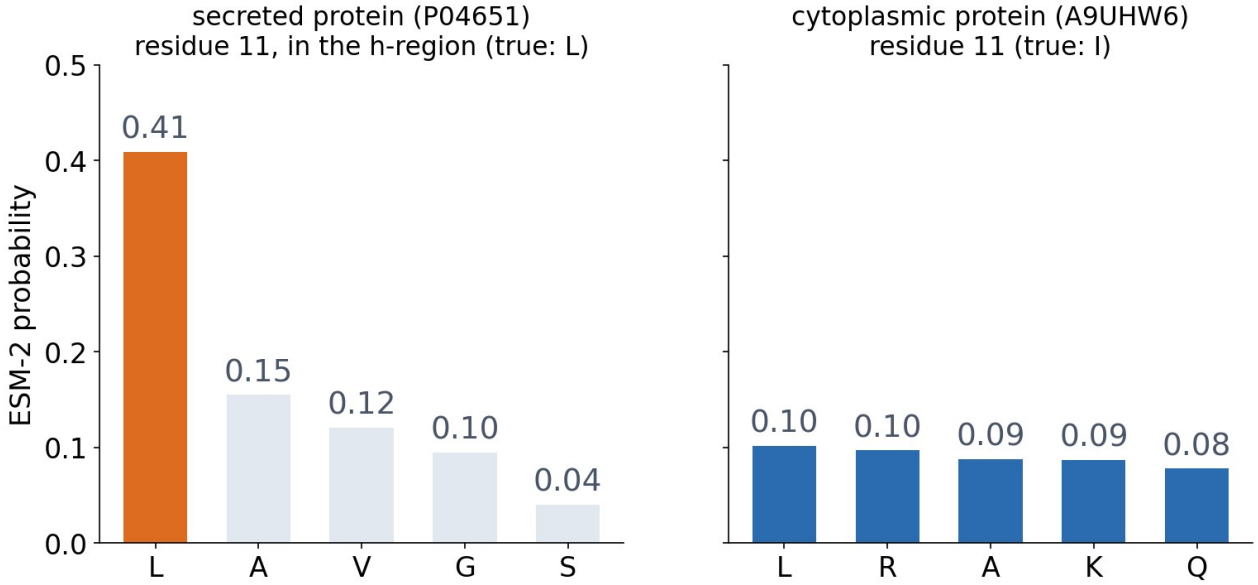


Attention was first added to recurrent translation models; the Transformer kept only the attention and processed all positions in parallel, which made large-scale pretraining practical.

Bahdanau et al. 2014; Vaswani et al. 2017; Radford et al. 2018; Devlin et al. 2018; Lin et al. 2023.

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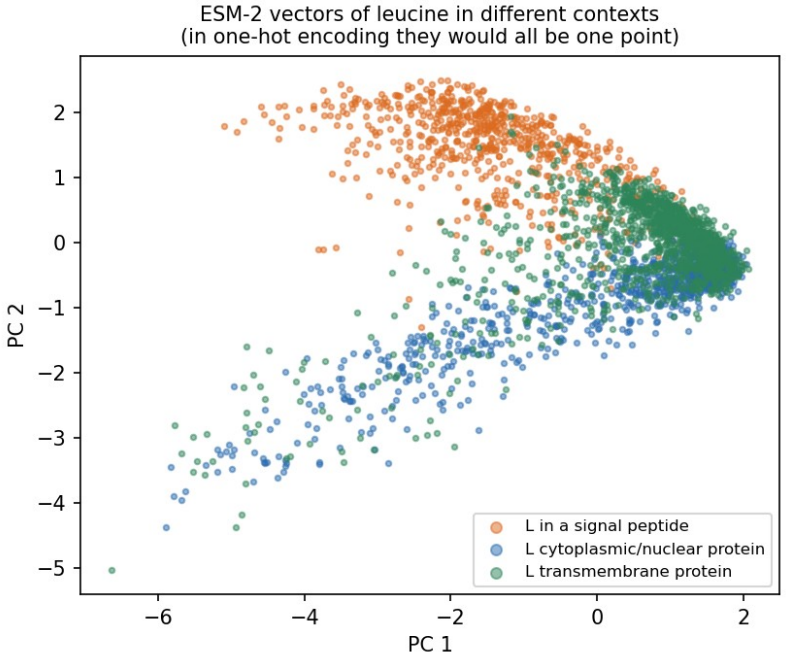
ESM-2 knows what a signal peptide looks like



- **h-region: L, 0.41**
- Cytoplasmic: no residue above 0.11
- Nobody told it about signal peptides

Notebook, Part 2: ESM-2 35M, residue 11 masked.

Same residue, different context, different vector

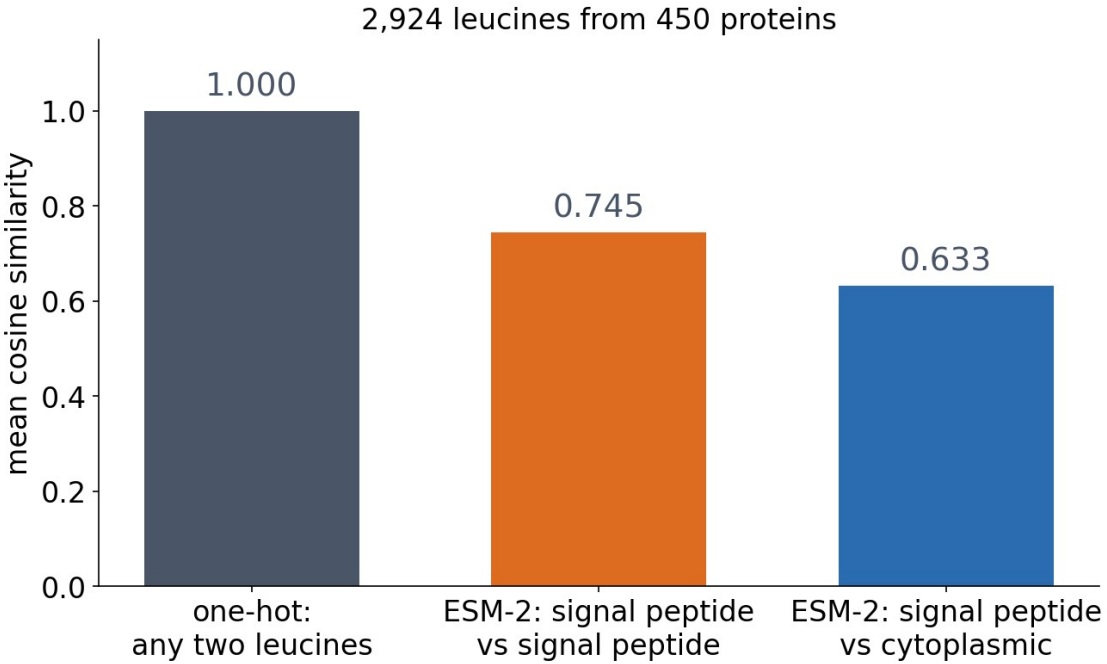


- **One-hot: every leucine the same**
- ESM-2: each computed from its surroundings
- **A contextual embedding**

Notebook: 2,924 leucines from the first 70 residues of 450 proteins.

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The vector carries the surroundings

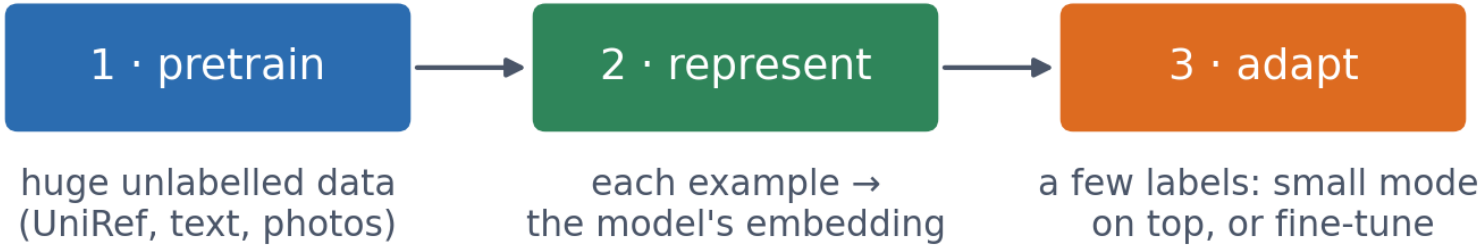


- **One-hot: cosine 1 for any pair**
- ESM-2: 0.745 within signal peptides
- 0.633 against cytoplasmic leucines

Notebook, Part 2.

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Pretrain once, reuse everywhere



met three times: DINOv2 on histology (Session 10) · ESM-2 · SignalP 6.0

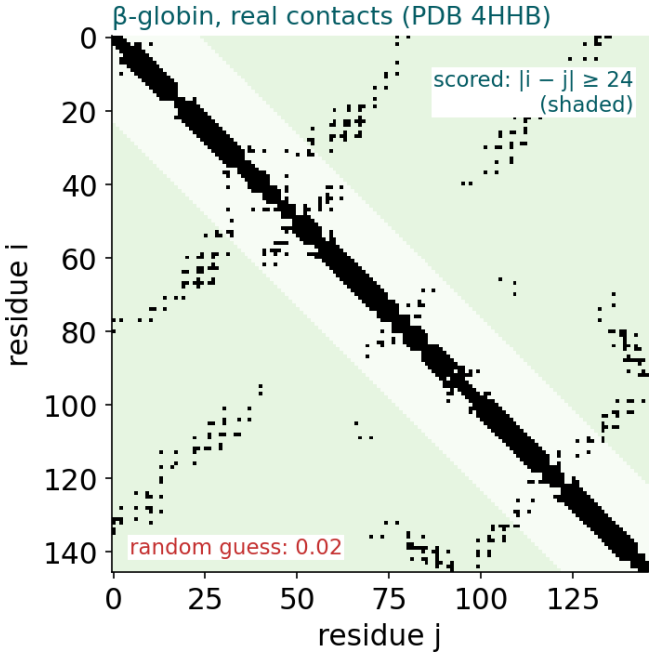
A foundation model helps most when labelled data are scarce, but it does not remove the need for an honest split.

Bommasani et al. (2021), arXiv:2108.07258.

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Does ESM-2 know β-globin's contacts?

Trained only on masked residues. Top-L/5 long-range precision from 8M to 650M parameters?



A About 0.9 at every size

B Near 0 at every size

C Near 0 at 8M, about 0.9 at 650M

D Highest for the smallest: big models overfit

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Does ESM-2 know β-globin's contacts?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Does ESM-2 know β-globin's contacts?

Fix it, run it, discuss

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Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session11-discussion-2.ipynb

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Does ESM-2 know β-globin's contacts?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

SciLifeLab

Does ESM-2 know β-globin's contacts?

Results, and who explains

Both rounds side by side.

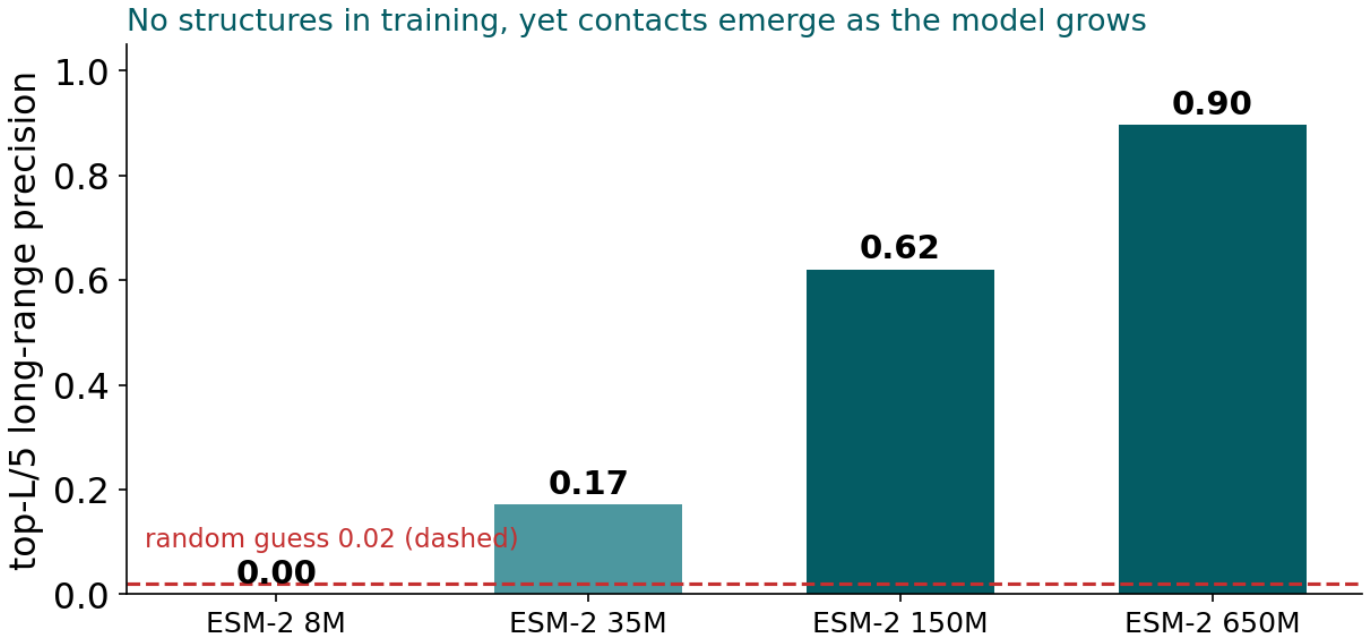
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session11-discussion-2.ipynb

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The answer: contacts emerge as the model grows



None of the models saw a structure. The smallest finds no long-range contacts; the 650M model gets nine of ten right. Residues in contact co-vary across a protein family, and a model large enough to fill in masked residues well learns which positions inform each other.

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BLOCK 2 DONE

Break

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BLOCK 3 · ABOUT 30 MIN

Pretraining changes biological prediction

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SignalP: thirty years in three steps



Each model type of the course appears: feed-forward networks on windows (Session 7) with hidden Markov models (Session 5), a deep recurrent network (Block 1), and a pretrained protein language model (Block 2).

Nielsen et al. 1997; Almagro Armenteros et al. 2019; Teufel et al. 2022; Nielsen et al. 2024.

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What pretraining changed

- **Rare types: predictable for the first time**
- No organism input needed
- Partitioning redone (GraphPart): Session 8

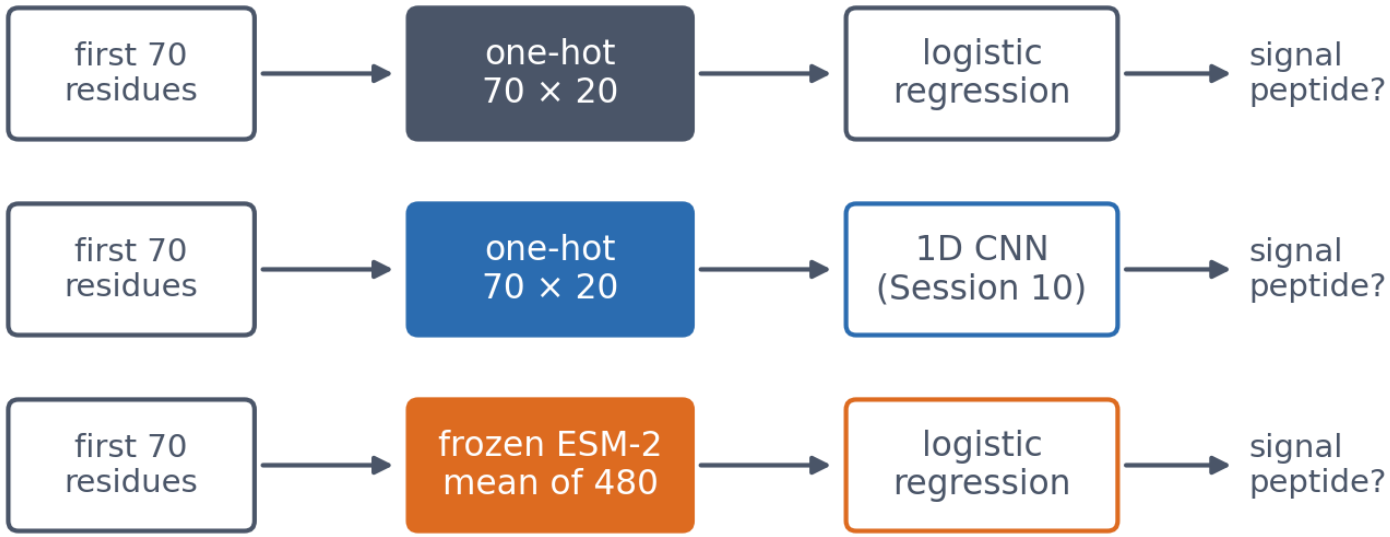
36 / 113
training sequences
of two rare types

< 60%
identity: 6.0 beats 5.0

Teufel et al. (2022), Nat Biotechnol 40:1023; Teufel et al. (2023).

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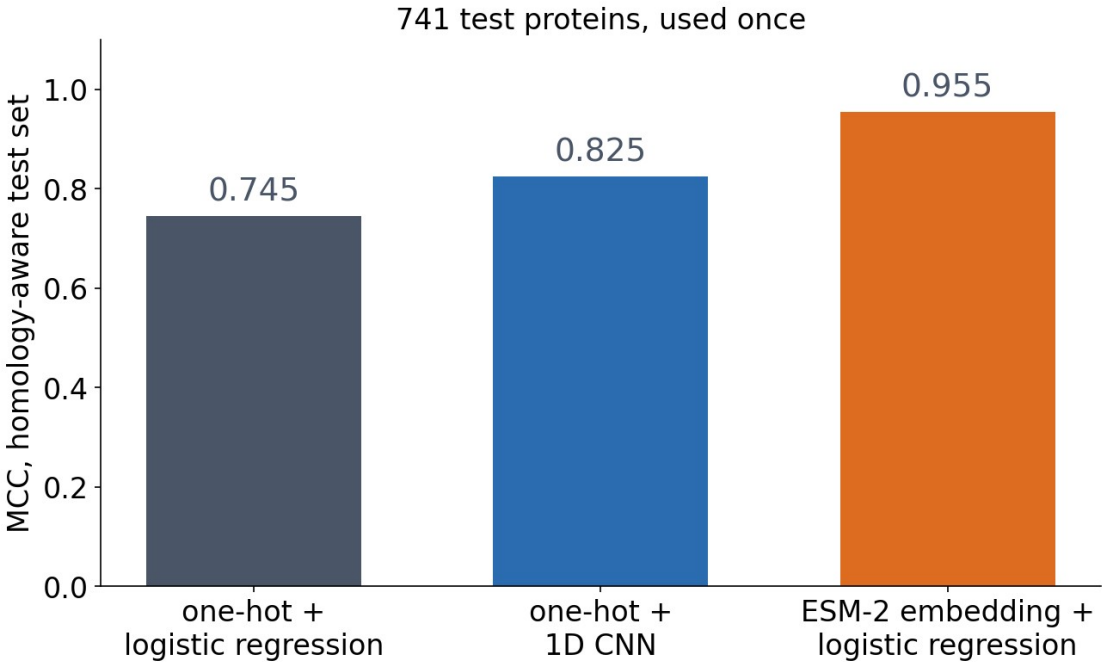
Does this protein have a signal peptide? Three classifiers



Same 4,910 proteins, first 70 residues. ESM-2 35M stands in for the larger ProtBERT; its residue vectors are averaged into 480 numbers and kept frozen.

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Frozen ESM-2 wins on the honest test

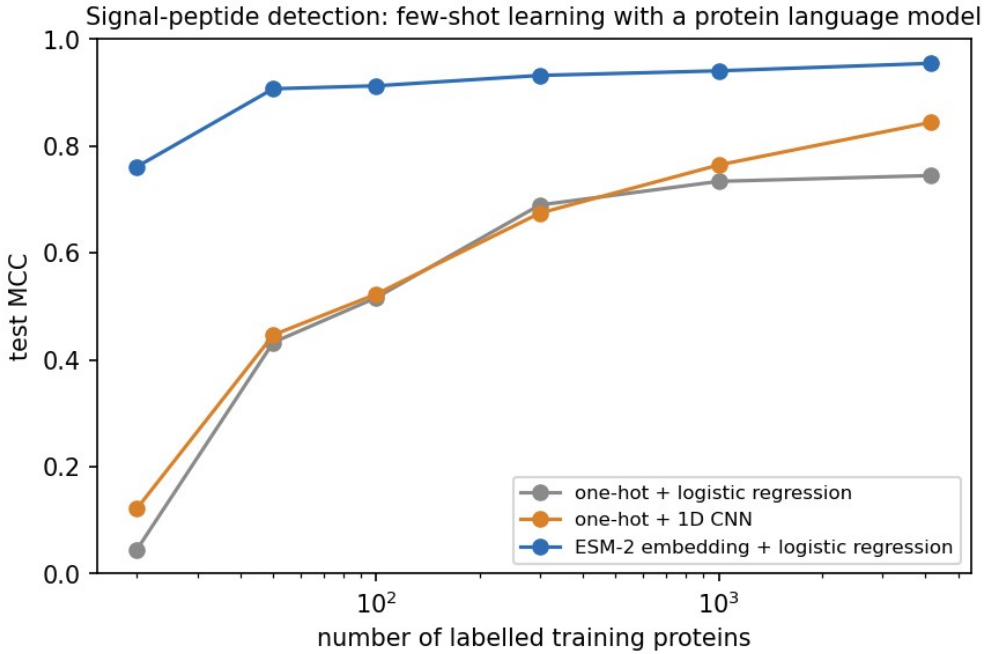


- **ESM-2: MCC 0.955**
- One-hot CNN: 0.825
- One-hot linear: 0.745

Notebook, Part 3; homology-aware split.

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Few labels: where pretraining pays off



- **20 proteins: 0.762 vs 0.045 / 0.122**
- 50 with ESM-2 (0.907) beat 4,169 with one-hot
- The labels only point at what the model already knows

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Gains where there is no close homolog

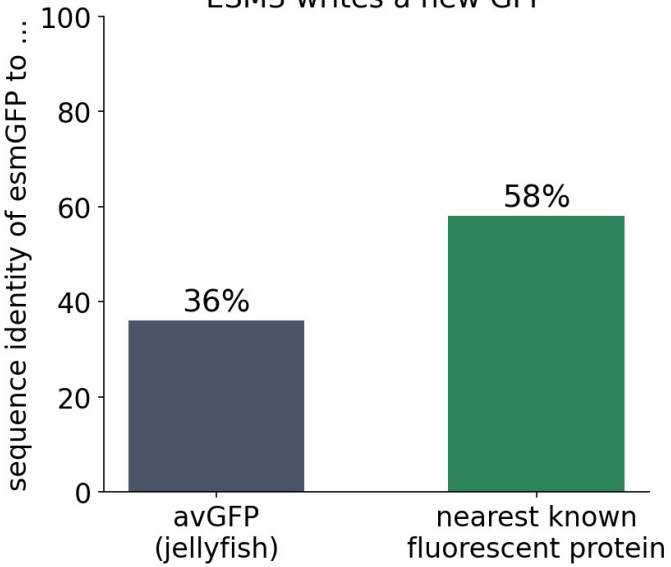


- **> 90% identity: all perfect**
- < 30%: one-hot 0.68-0.77, ESM-2 0.96
- The SignalP 6.0 pattern
- Random split: a diagnostic only

Notebook, Part 3: random 80/20 split, identity to the most similar training protein by MMseqs2.

From reading sequences to writing them

ESM3 writes a new GFP



Evo 2 reads and writes DNA

40 billion

parameters

~9 trillion

DNA bases, all domains of life

1 million

bases of context

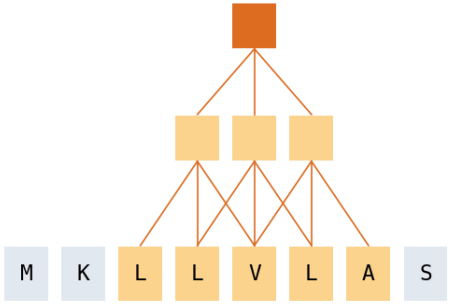
- ESM3 designed a new, glowing GFP
- Evo 2: genomes, all domains of life
- Next: virtual cells (Alpha Cell, Session 10)
- **Does a protein language model understand biology, or only model sequences?**

Hayes et al. (2025), Science 387:850; Brixi et al. (2026), Nature (Evo 2, Arc Institute); scilifelab.se/alpha-cell.

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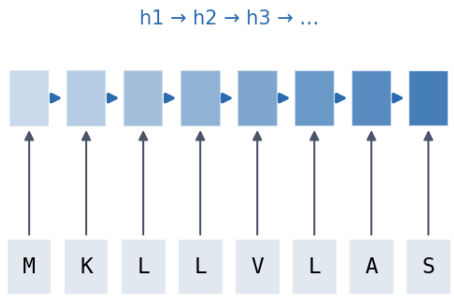
Three ways to use context, two ways to learn a representation

CNN: local context grows layer by layer



a unit sees only its receptive field

RNN: a hidden state carries context



context must pass through every step in between

Attention: every position consults every other



one step, any distance; weights say how much each matters

CNN: local context grows layer by layer. RNN: context carried step by step. Attention: any position, directly. A supervised model learns its representation from labels; a protein language model learns it first from millions of unlabelled sequences, so that a few labels go a long way.

See course.bioinfo.se/session11.html, 'What's next'.

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Does a signal-peptide detector transfer across kingdoms?

Trained on eukaryotes, tested on bacteria: one-hot residues against frozen ESM-2 embeddings.



`features 1:` one-hot, first 70 residues

`features 2:` frozen ESM-2 embeddings

same logistic regression on each

A Both keep nearly all their performance

B One-hot does better

C ESM-2 keeps most; one-hot loses much more

D Both fall to near zero

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Does a signal-peptide detector transfer across kingdoms?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Does a signal-peptide detector transfer across kingdoms?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour



The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session11-discussion-3.ipynb

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Does a signal-peptide detector transfer across kingdoms?

Answer again

Same question, after the discussion.



quiz.bioinfo.se/live

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Does a signal-peptide detector transfer across kingdoms?

Results, and who explains

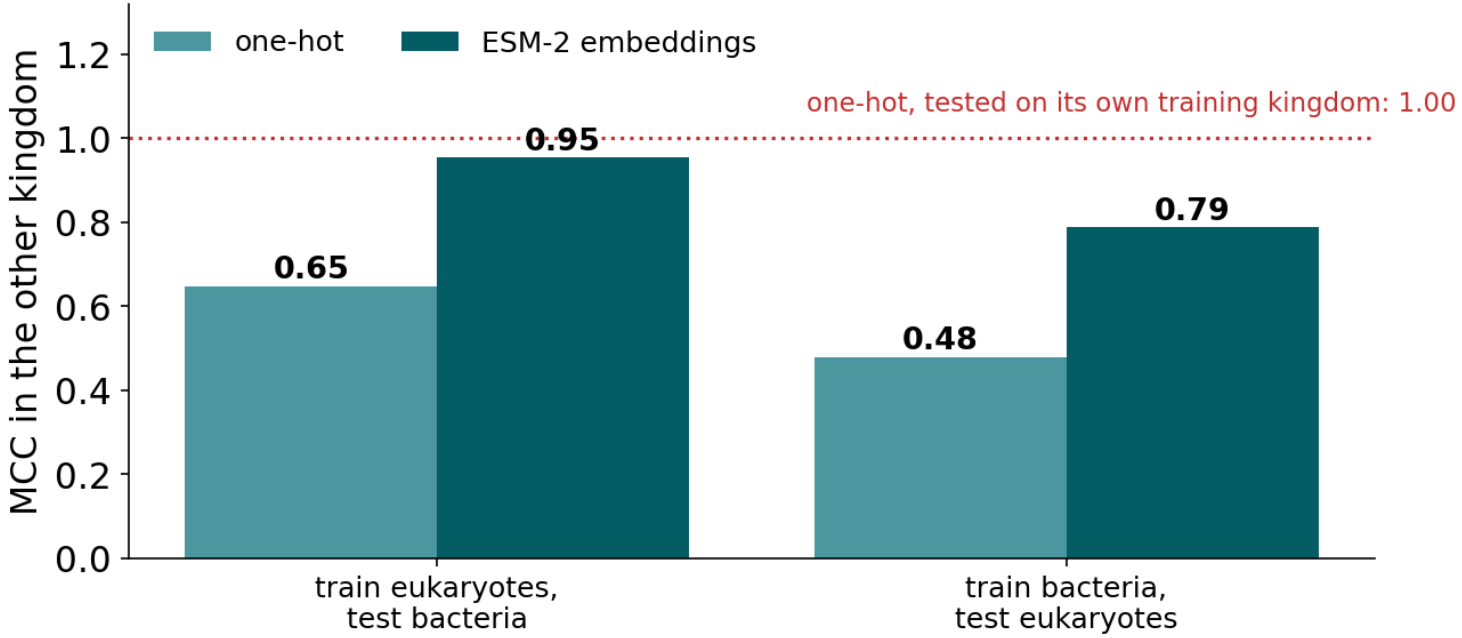
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: ESM-2 transfers, one-hot much less



On its own training kingdom the one-hot model looks perfect (MCC 1.00), but in the other kingdom it falls to 0.65 and 0.48. Frozen ESM-2 embeddings keep 0.95 and 0.79: the representation was learnt from unlabelled sequences of all kingdoms, which supports SignalP 6.0's choice.

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In up to three words: what was the hardest part of today?

Remembering context

RNN, LSTM, bidirectional

Learning without labels

language models, attention, Transformer

Pretraining pays off

SignalP, ESM-2 embeddings

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

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Session 12 next

AlphaFold: attention and pair representations reason about 3D structure
Where long-range structural information comes from, and what a confident model means